

1. The melting point of TiCl_4 (-24°C) lies below those of TiF_4 (284°C) and TiBr_4 (38°C). Explain why by considering the covalent-ionic nature of these compounds and the intermolecular forces in each case.
2. Show how the glycinate ion ($\text{H}_2\text{N}-\text{CH}_2-\text{COO}^-$) can act as a bidentate ligand. (Draw a Lewis diagram if necessary.) Which atoms in the glycinate ion will bind to a metal ion?
3. Give the chemical formula of each of the following compounds:
 - (a) Silver hexacyanoferrate(II)
 - (b) Potassium tetrakisothiocyanatocobaltate(II)
 - (c) Sodium hexafluorovanadate(III)
 - (d) Potassium trioxalatochromate(III)
4. Draw the structures of all possible isomers for the following complexes. Indicate which isomers are enantiomer pairs.
 - (a) Bromochloro(ethylenediamine)platinum(II) (square-planar)
 - (b) Tetra-amminedichloroiron(III) ion (octahedral)
 - (c) Amminechlorobis(ethylenediamine)iron(III) ion (octahedral)
5. Platinum(IV) forms octahedral complexes. Sketch the structures of all the distinct isomers of $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2\text{F}_2]$, indicating which pairs of structures are mirror images of each other